



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering
Academic Year: 2023- 2024 (Odd Semester)

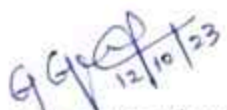
INNOVATIVE TEACHING METHOD

Degree, Semester & Branch: III Semester B.E. ECE B

Course Code & Title: EC3354 Signals and Systems

Name of the Faculty member: Mrs.G.Gnana Priya

SL.No.	Topic(s)	Activity	Reference(s)
UNIT I - CLASSIFICATION OF SIGNALS AND SYSTEMS			
1.	Classification of systems - Causal & Non-causal, Stable & Unstable, Revision-Unit I	Mind Map	Allan V. Oppenheim, S. Willsky and S. H. Nawab, "Signals and Systems", Pearson Education, 2007.
			


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Innovative Practice Description

- **Unit / Topic:** Unit II / Analysis of Continuous Time Signals – Laplace Transform Properties
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO7
- **Activity Chosen:** Flipped Classroom
- **Justification:**

The Unit II consists of properties of Laplace Transform. The concept of Laplace Transform is very important. The Fourier transform was taught in the previous topic and I explained the relation between Fourier transform and Laplace transform in the class which helped them to prepare and execute the Flipped Classroom activity. From their presentation I can observe the understanding level of the students.

- **Time Allotted for the Activity:** 50 minutes
- **Details of the Implementation:**

The materials related to the activity were given to the students for learning. The activity was implemented by forming teams. The team forming choice was given to the students with the condition that a team should consist of mixer of advanced learners, average learners and slow learners. In such a way the class strength was divided into 10 teams. Each team prepared and presented the properties of Laplace transform as follows:

- ↓ Linearity Property
- ↓ Transform of Derivatives
- ↓ Scaling Property
- ↓ Time Shifting property
- ↓ Frequency Shifting property
- ↓ Differentiation in the s-domain
- ↓ Time convolution
- ↓ Frequency convolution
- ↓ Initial value theorem
- ↓ Final value theorem

CO – PO / PSO mapping:

CO	PO1	PO8	PO9	PO10	PSO1
CO2	2	2	3	2	2

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO1	PO8	PO9	PO10	PSO1
	3	2	3	2	2
Justification for correlation	Analysis of CT signals requires the knowledge of mathematical and engineering fundamentals such as integration, differentiation, and transforms. So, the course outcome is highly correlated.	The course outcome is moderately correlated, since the Flipped Classroom activity involves presentation which applies moral & ethical principles while collecting the materials related to the topic.	A collaborative activity (flipped classroom) is planned where the students work as a member of a team to achieve the goal. So, the course outcome is highly correlated.	Through flipped classroom activity the students' listen to, comprehend information, instructions and viewpoints of others. So, the course outcome is mapped at level 2.	The students will be able to innovate new ideas to solve existing/novel problems in the analysis of CT signals. So, the course outcome is mapped at level 2.

• Images / Screenshot of the practice:





Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

The following points were observed based on the feedback got from the students.

- ✓ The students felt that the self-learning capability improved because of this activity.
- ✓ The presentation skills also increased while learning the concept by themselves and explaining to the class.
- ✓ The communication skills of the students improved while discussing with their team members and then presenting to the class.

❖ Benefit of the practice:

- ✓ The main benefit of the flipped class activity is each group will learn one property and the presentation helps the other members also to know about the other properties.
- ✓ The students effectively involved in the activity by discussing with their team members.
- ✓ The understanding level and learning level of the students also increased.
- ✓ As the team forming choice was given to the students they actively participated.

❖ Challenges faced in implementation:

- The biggest challenge was making the students to learn the materials. Each student has to read the material posted in the canvas and then they have to prepare.
- Very few students were not actively participated in the activity. Making all the students to participate was also challenging.
- The activity was planned for one hour but it took extra time than the allotted time to complete the activity.

References:

- ❖ Allan V. Oppenheim, S. Wilsky and S. H. Nawab, "Signals and Systems", Pearson Education, 2007.
- ❖ A. Nagoor Kani, "Signals and Systems", Tata McGraw Hill Education Private Limited, Second Edition, 2010.


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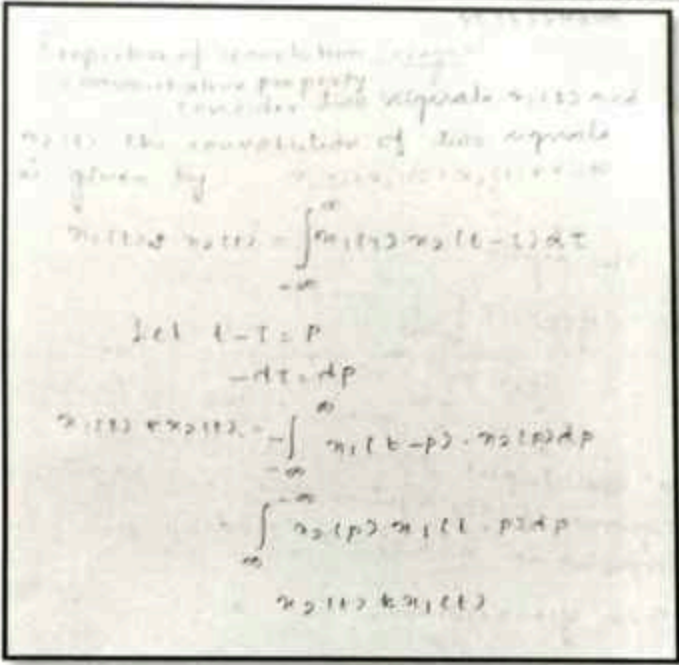
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Sl.No.	Topic(s)	Activity	Reference(s)
UNIT III LINEAR TIME INVARIANT – CONTINUOUS TIME SYSTEMS			
I.	Convolution integrals - Properties	Minute Paper	R.E.Zeimer, W.H.Tranter and R.D.Fannin, "Signals & Systems - Continuous and Discrete", Pearson, 2007.
 Properties of convolution integrals Commutative property Consider two separate signals $x_1(t)$ and $x_2(t)$. The convolution of two signals is given by $x_1(t) * x_2(t) = \int_{-\infty}^{\infty} x_1(\tau) x_2(t - \tau) d\tau$ Let $t - \tau = p$ $-d\tau = dp$ $x_1(t) * x_2(t) = \int_{-\infty}^{\infty} x_1(t - p) \cdot x_2(p) dp$ $= \int_{-\infty}^{\infty} x_2(p) \cdot x_1(t - p) dp$ $= x_2(t) * x_1(t)$			


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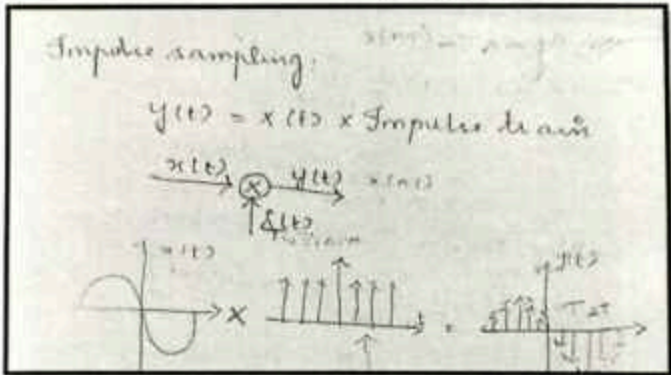
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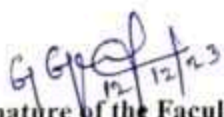
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Sl.No.	Topic(s)	Activity	Reference(s)
UNIT IV ANALYSIS OF DISCRETE TIME SIGNALS			
1.	Baseband signal Sampling	Sketch Noting	Allan V. Oppenheim, S. Willsky and S. H. Nawab, "Signals and Systems", Pearson Education, 2007.
			


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Sl.No.	Topic(s)	Activity	Reference(s)
UNIT V LINEAR TIME INVARIANT - DISCRETE TIME SYSTEMS			
1.	Difference Equation- Natural & Forced Response, Total Response	Sorting strips	Allan V. Oppenheim, S. Willsky and S. H. Nawab, "Signals and Systems", Pearson Education, 2007.
			


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